Turing Test During Olympics and Minsky's Anger – – Some Pros and Cons of Dialog Systems Competition

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In this paper we introduce our considerations about probably the oldest and the least popular and most difficult part of competitive natural language processing - the dialog systems contest and describe problems of its most famous representative - the Loebner Prize. We begin with the description of the contest, its presuppositions, participation system and judging method then explain the problems which emerged due to the technical failures and unclear approach to evaluation. In the next part we will propose a different approach and technical solutions for the dialog systems tournament. In the end, for rousing a discussion about competitive NLP we also suggest organizing an NLP Olympics where Turing test-like competition could become a place to combine information retrieval, summarization and question answering systems contests.

Keywords: Competitive Natural Language Processing, Dialog Systems, Turing test

オリンピックでのチューリング・テストとミンスキーの怒り – – 対話システムコンテストについての賛否両論

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本稿では最古かつ多くの問題点が指摘されている対話システムコンテスト「Loebner賞」を紹介し、その競争的な「評価型自然言語処理」の一部の問題について考察する。コンテスト、その前提、参加システム、評価方法の説明の後、曖昧なアプローチ及び技術的なトラブルによって起きた問題がどのように評価へ影響を及ぼしたか説明する。次に、対話システム・トーナメントの異なるアプローチと技術的なソリューションを提案する。最後に、評価型自然言語処理についてディスカッションのきっかけとして、情報抽出、自動要約や質問応答などを含むチューリングテスト風の競技になる可能性を示唆するNLPオリンピックの実施を提案する。

キーワード：評価型自然言語処理、対話システム、チューリングのテスト

1 INTRODUCTION

Famous British mathematician Alan Turing when asked if the computers can think used to reply that the question is meaningless. He claimed that a more well-defined question is needed and he proposed a test, which was inspired by a party game known as the "Imitation Game" – a man and a woman go into separate rooms, and guests try to tell them apart by writing a series of questions and reading the typewritten answers sent back [1]. In this game, both the man and the woman aim to convince the guests that they are the woman. This idea is nowadays being used as an intelligence test for computers known as "Turing test" during an annual contest called "Loebner Prize". With growing number of evaluation-based workshops in different areas of Natural Language Processing (Question Answering, Text Summarization, Text Retrieval, Machine Translation) on the one side, there is only Dr. Loebner's competition where the freely talking (chatting) systems can be compared. Although all controversies in the AI field which aroused around the Prize and relatively low acceptance for so called "chatterbots" we will argue in this paper that this kind of challenge, after several modifications, could become an opportunity for joining AI specialists in cooperative groups where the exchange of ideas and elaborate standards for future team working. We claim that such an event could also gain a better publicity for AI and NLP in media and among other scientists as many of them consider this field as divided into too many pieces of very particular and limited tasks.

1.1 The History of Loebner Prize

The Loebner Prize is an annual competition that awards prizes to one chatterbot which is considered the most humanlike for that year. The Prize was established in 1990 by Dr. Hugh Loebner in conjunction with The Cambridge Center for Behavioral Studies and the first competition was held in 1991. Sponsored by public and private money (in the beginning by National Science Foundation and Sloan

Foundation), Dr. Loebner had set following prizes:

- USD 2,000 for the most human seeming of all bots for that year - awarded every year
- USD 25,000 for the first bot that judges cannot distinguish from a real human in a text-only based Turing Test (awarded once only)
- USD 100,000 to the first bot that judges cannot distinguish from a real human in a Turing Test that includes deciphering and understanding text, visual, auditory (and tactile?) input; the Loebner Prize dissolves once the USD100,000 prize is won.

1.2 Rules of the Competition

There are Entries (programs), Judges and so called Confederates who are evaluated by the Judges on the same basis as Entries. Programs have to be stand-alone units with communication programs to connect to other computers. If there is more than four and less than eight competitors, the Judges evaluate systems in real-time tests and choose four best Entries. If there is more than eight competitors, Entries are prescreened (evaluation of replies to the same set of utterances). Conversations must be always started by the Entry's self-introduction in given form. There are no other restrictions on conversations or topics. Scoring is described as follows (for 2005):

- The Final Competition is scored by comparing pairs.
- Each Judge evaluate four pairs of terminal. One terminal of each pair is controlled by an Entry. The other terminal will be the matching communications program supplied along with the Entry. That program will be controlled by a Confederate.
- The task of each Judge will be to apportion 100 points between the two terminal based upon how "human" they seem. The judgment is relative, and ties are forbidden; one of the two terminal must be awarded at least 51 points. No judge may use the same apportionment twice.
- (a) Each Entry to be compared with every Confederate; (b) each Judge to evaluate every Entry, (c) each Judge to evaluate every Confederate.
- The Entry with the highest column total is declared the winner.

The Silver Medal and USD25,000 will be awarded if the score is "2" or higher as Loebner believes that an Entry which has been independently rated by two different Judges as "more human" than two different Confederates in direct paired comparisons deserves the Silver Medal.

1.3 Loebner's Reason

The most problematic for the task were the premises of the founder, which appeared not really scientific. This was one of the reasons that noble AI researchers have abandoned Dr. Loebner with his idea but, what cannot be negated, the results of the first tests were so bad that also made the scientists and journalists lose their interest. Oppositely, many famous personas of the field felt offended as the victory of the ELIZA [4] clone was commented with a suggestion that 25 years of research brought no fruits at all[6]. One of them was prof. Marvin Minsky who personally attacked the Loebner Prize in 1995 at Usenet's comp.ai.philosophy. However, even if the suggestion hurt feelings, it showed how the publicity regards computer science and aroused a debate on how the intelligence could be measured, if can it be measured at all, and what is the intelligence itself. The concept might have been correct but it seems that people involved are too eccentric and there is too many dark clouds hanging over their heads to make the competition useful and fair. We point out some other problems below.

2 PROBLEMS

2.1 Why ELIZA Won the First Prize Ever?

Weintraub (ELIZA system) and Colby (Parry system[5]) came to the competition for showing their psychology-based approach and their methods were winning. In the beginning the Turing test was limited - the topic and tenure were established. This was easy to Weintraub who has chosen "whimsical conversation" for his bot. On the other hand a confederate who was a devoted Shakespeare fan was thought to be a machine as the judges could not believe that human knows so much about the writer. The rules were finally changed but after the people proposing those changes had already quit the Loebner Prize.

2.2 Technical Problems Influence the Results

Another problems which appeared during the first competitions were the technical failures. Half of the computers prepared for the contest did not want to connect and the

number of Confederates had to be decreased. It made the scoring system even more inadequate and not really clear judging system became severely criticized [3]. Another problem was with the typing speed - the Entries (programs) were giving slow responses to simulate human but some Confederates were just too slow and it seemed not everyone understood properly the organizer guidelines - many tricky questions were made what had nothing to do with natural conversation aimed by Loebner's team. It became an illegal behavior after changing the contest rules.

3 ANOTHER APPROACH

3.1 To Discuss the Experiences

As we can see with this controversial example, organizing an evaluating event for free conversation systems we can meet the wall of different kind of problems – from technical to personal as Turing test, though very famous, is not regarded by all researchers of the field [2] as the best and "the one and only" method of evaluating such systems. As the term "AI" itself has devaluated, "decision making intelligent systems" school (for example Minsky) is not likely to take part in workshops where the stress is put on the opposite school where imitating a human is the main point. In our opinion, as in sports, if some organizations (it is better if there is more than one entity) make fair rules and disciplines for everyone and organizes multi-contest giving everyone an opportunity to approve themselves and to discuss – probably more participants would take part in such event. Especially because, as mentioned in the introduction, there are already big evaluation events organized around the world.

3.2 NLP Olympics

We titled the idea which we are going to introduce during the Takamatsu Workshop an "NLP Olympics" but we imagine such event rather as "NLP *Kohaku* Tournament"[1]. For encouraging the cooperation between NLP teams we propose annual random establishment of participating groups or persons for building a modular NLP system which (among other disciplines) could also start in the Turing test. Such group could have experts from QA, summarization, paraphrasing fields and also work together with cognitive science or psychology specialists, linguists, etc. In one discipline the original Turing test could be replaced with two computer programs talking about, for example, random text to show how the NLP systems can cooperate with each other. "Human-like" (this criteria should be discussed) degree could be evaluated electronically by many people watching the contest on the one side, and by the experts on the other side. Machine-translated output could be evaluated online by native judges. Several subdisciplines could be also proposed while the "Olympics" evolve.

[1] "Kohaku" is famous Japanese New Year's singing contest where all the participants are divided into two groups - Reds and Whites

4 CONCLUSIONS

We briefly described the organization and problems of the annual Turing test competition called a Loebner Prize. We explained the rules and the judging system and pointed out what made the test controversial and why it was abandoned by the scientists. Then we suggested our vision of using Turing test for combining the AI (in this case NLP) research groups into groups which could compete on the fair-play basis. We leave the discussion open as there are threats as doping in real sports and leave questions as "is making the sources open enough to avoid zillion dollar tuning by the biggest companies?" unanswered for provoking the discussion.

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